Living Costs and Food Survey 2013 – Case Study: Is There a Relationship Between Occupational Class, Tenure Type, Number of Adults, Number of Children and Household Expenditure?

Data Analysis Course

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1. INTRODUCTION

This case study examines the relationship between household expenditure and a set of socio-economic and demographic characteristics using data from the *Living Costs and Food Survey (LCF) 2013 teaching dataset (UK Data Service, 2014)*. The objective of the analysis is to investigate whether occupational class, tenure type, number of adults, and number of children in a household are associated with differences in total household expenditure.

Understanding how household characteristics influence spending behaviour is important for economic research, public policy, and social analysis. Household expenditure is a key indicator of living standards and cost of living, and it may vary depending on household composition, employment status, and housing conditions (Office for National Statistics, 2014). Differences in expenditure across socio-economic groups may reflect inequalities in income, needs, and consumption patterns.

To address the research question, statistical analysis and data manipulation were carried out using **Python** and the **pandas library** (Pandas Development Team, 2023). A multiple linear regression model was estimated using Ordinary Least Squares (OLS) via the statsmodels library. The model was then used to evaluate the relationship between total household expenditure and the selected explanatory variables. In addition to the baseline model using raw expenditure, a logarithmic transformation of expenditure was estimated in order to improve the distributional properties of the model and to assess the robustness of the results.

Diagnostic tests and graphical analysis were performed to evaluate the assumptions of the regression model, including normality of residuals, homoscedasticity, independence of observations, and multicollinearity. Where necessary, heteroskedasticity-robust standard errors were used to ensure valid statistical inference. The results of the regression analysis, together with complementary descriptive statistics and diagnostic tests, were used to determine whether household expenditure is significantly related to occupational class, tenure type, number of adults, and number of children.

2. DESCRIPTION OF THE SURVEY

The data used in this study come from the *Living Costs and Food Survey (LCF) 2013*, a large-scale household survey conducted in the United Kingdom (Office for National Statistics, 2014). Participation in the survey is voluntary. The original survey collects detailed information on household expenditure, income, and living conditions, and is designed to provide a comprehensive picture of spending patterns across different types of households.

For the purposes of this case study, a reduced teaching dataset derived from the original survey was utilized (UK Data Service, 2014). This dataset contains fewer variables but preserves the structure and main characteristics of the full survey, making it suitable for statistical analysis and teaching applications.

The Living Costs and Food Survey is one of the main sources of information on household expenditure in the United Kingdom. The data are used for the construction of the Retail Price Index, National Accounts estimates, analysis of taxes and benefits, and research on nutrition, poverty, and living standards. The survey is carried out by the Office for National Statistics (ONS) in Great Britain and by the Northern Ireland Statistics and Research Agency (NISRA) in Northern Ireland.

The survey consists of three main components: a household questionnaire, an individual questionnaire, and a two-week expenditure diary. The household questionnaire collects information about household composition, housing conditions, and major purchases. The individual questionnaire gathers information on employment status, income, and benefits for each household member. The expenditure diary records detailed daily spending over a two-week period and forms the basis for total household expenditure.

The teaching dataset used in this analysis contains 5,144 households and 19 variables. Among these households, 151 observations come from Northern Ireland. The dataset includes information on total household expenditure, occupational class, housing tenure, number of adults, and number of children, which are the variables used in the regression analysis.

3. RESEARCH QUESTION AND MOTIVATION

The research question examined in this case study is:

Is there a relationship between occupational class, tenure type, number of adults, number of children, and total household expenditure?

This question is of particular interest because household expenditure is influenced by both socio-economic status and household composition. Occupational class is commonly used as an indicator of income level and employment status, tenure type reflects housing conditions and financial commitments, while the number of adults and children affect household needs and consumption requirements. Differences in these characteristics are expected to lead to systematic differences in spending behaviour.

By examining the relationship between these factors, it becomes possible to identify patterns in household spending. The results of this analysis aim to explain how different types of households allocate their spending and which factors are most strongly associated with higher or lower expenditure. Analyzing these relationships can help businesses better understand consumer behavior and target or create products or services more effectively. At the same time, governments can use this information to design better economic policies, social support programs, and public services that meet the needs of different household groups.

To answer the research question, the relationship between household expenditure and the selected explanatory variables is examined using multiple linear regression estimated by Ordinary Least Squares (OLS). This approach allows the effect of several variables to be analysed simultaneously, making it possible to determine whether each factor has a statistically significant association with expenditure while controlling for the others.

4. DATA AND VARIABLES

To answer the research question, five variables from the teaching dataset were selected. The dependent variable is total household expenditure, while the independent variables describe socio-economic status and household composition. These variables were chosen because they are expected to influence household spending and are commonly used in economic analyses of living standards.

The variables used in the analysis are described below.

4.1 Dependent Variable: Household Expenditure

This variable represents total household expenditure and is measured in pounds per week (£). It is a continuous numerical variable and is used as the dependent variable in the regression analysis. Because expenditure is positively skewed, a logarithmic transformation of expenditure was also used in an alternative specification in order to improve the distributional properties of the model. The log-transformed variable serves as the primary dependent variable in the final model to ensure normality of residuals and to allow for percentage-based interpretations.

4.2 Independent Variables

Socio-Economic Status: Occupational Class

This variable describes the occupational class of the household reference person. It is a categorical variable with several categories representing different socio-economic groups, including higher occupations, intermediate occupations, routine and manual occupations, unemployed or not stated. In the regression model, this variable is treated as a categorical factor and included using dummy variables, with one category used as the reference group.

Tenure type

This variable describes the housing tenure of the household. It is a categorical variable with three categories: owned, public rented, and private rented. These categories do not have a natural numerical order.

In the regression analysis, tenure type is included as a categorical factor using dummy variables, allowing the model to estimate differences in expenditure relative to the reference category.

Number of adults

This variable represents the number of adults living in the household. In the original dataset, the values were stored as text categories (for example, “1 adult”, “2 adults”, “3 adults”, “4 and more adults”).
For the purposes of the regression analysis, this variable was treated as a categorical variable rather than a numerical one and included in the model using indicator (dummy) variables. This approach allows the model to capture non-linear differences in expenditure across household sizes.

Number of children

This variable indicates the number of children living in the household. It is a categorical variable with the categories: no children, one child, and two or more children.
The variable is included in the regression model as a categorical factor using dummy variables, so that differences in expenditure between households with different numbers of children can be estimated without assuming a strictly linear effect.

Variable	Type	Role	Transformation
Expenditure	Continuous	Dependent	Log-transformed (ln)
Occupational Class	Categorical	Independent	Dummy coded

Tenure Type	Nominal	Independent	Dummy coded
No. of Adults	Categorical	Independent	Dummy coded
No. of Children	Ordinal	Independent	Dummy coded

Table 1. Variable types

5. DESCRIPTIVE STATISTICS

Before conducting the regression analysis, the distribution and characteristics of the dataset were examined to understand the underlying patterns of household expenditure and composition.

Household Expenditure Summary

The dependent variable exhibits a wide range of values, reflecting diverse consumption patterns across the sample.

Statistic	Value (£)
Observations (n)	5,144
Mean	479.76
Median	419.9
Std. Deviation	292.37
Minimum	30.53
Maximum	1,175.00

Table 2. Statistical Values

Descriptive statistics indicate a mean of £479.76 and a median of £419.90. The difference between the mean and the median suggests a positively skewed distribution, consistent with higher expenditure values in the upper tail. The standard deviation is relatively large (£292.37), indicating substantial variability in household spending across the sample. Expenditure ranges

from a minimum of £30.53 to a maximum of £1,175.00, highlighting the wide dispersion in consumption levels.

Occupational class

In terms of occupational class, the largest group in the sample is 'Not classified for other reasons' (1,619 households), followed closely by 'Higher managerial, administrative and professional occupations' (1,589 households). 'Routine and manual occupations' account for 1,045 households, while 'Intermediate occupations' represent a smaller proportion (691 households). The smallest group consists of households where individuals have never worked, are long-term unemployed, or where occupation is not stated (200 households).

Tenure type

Regarding housing tenure, the majority of households are owner-occupied (3,466), followed by public rented accommodation (880) and private rented accommodation (798). This distribution suggests that home ownership is the dominant tenure type within the sample.

Household composition (adults)

The distribution of the number of adults shows that most households consist of two adults (2,826), followed by single-adult households (1,729). Households with three adults (422) and those with four or more adults (167) are considerably less common.

Household composition (children)

With respect to the number of children, most households have no children (3,605). Households with two or more children account for 885 observations, while those with one child represent 654 households. This indicates that childless households are the most prevalent in the dataset.

Overall, the descriptive statistics and category distributions reveal substantial variation in household expenditure and highlight the diversity of socio-economic and demographic characteristics within the sample, providing a solid basis for further analysis.

6. DATA PREPARATION AND TRANSFORMATIONS

Before performing the statistical analysis, the dataset was prepared in Python using the *pandas* library (Pandas Development Team, 2023). Only the variables required to answer the research question were selected from the full teaching dataset in order to simplify the analysis and focus on relevant household characteristics.

The variable names in the original dataset were coded using short labels, and were renamed to more descriptive names to improve readability and make the code easier to interpret. In particular, the variables representing occupational class, tenure type, number of adults, number of children, and total household expenditure were renamed using meaningful identifiers.

Some variables required transformation before they could be used in the regression model. The variable representing the number of adults was originally stored as text categories (for example, “1 adult”, “2 adults”, “3 adults”, “4 and more adults”). The numeric value was extracted from the text and the variable was converted into a categorical variable with clearly defined levels, so that it could be included in the regression model as dummy variables.

The number of children variable was kept as a categorical variable with three levels: no children, one child, and two or more children. This classification reflects the structure of the teaching dataset and allows the regression model to estimate differences in expenditure between household types without assuming a strict linear effect.

Because the expenditure variable is positively skewed, a logarithmic transformation was created as *log_expenditure = ln(expenditure)*. The log-transformed variable was used in an alternative regression specification in order to improve the distribution of residuals and to better satisfy the assumptions of the Ordinary Least Squares (OLS) model.

After these transformations, the dataset was ready for formal modeling using the **statsmodels** library (Seabold and Perktold, 2010).

7. GRAPHICAL ANALYSIS

To visualize the underlying patterns in the data, exploratory visualizations were constructed to understand the distribution of the variables and the relationships between them. These plots serve to validate the relationships suggested by the descriptive statistics and to inspect the data for non-linearity and outliers. Boxplots were used to compare household expenditure across occupational classes and tenure types. Additional plots examined how expenditure varies with the number of adults and children in the household. Together, these visualizations provide an initial view of the patterns that were later assessed more formally through regression analysis.

7.1 Distribution of Household Expenditure

As shown in **Figure 1**, household expenditure exhibits a pronounced right-skew (positive skew). The mean expenditure (£479.76) exceeds the median (£419.90), indicating that a small number of high-spending households increase the average. The histogram of raw expenditure shows most households clustered at lower spending levels, with a noticeable spike at the upper bound, suggesting potential top-coding. This concentration of extreme values highlights the risk that a raw linear model could be influenced by outliers.

Such skewness is common in expenditure data and raises concerns regarding the normality assumption required for Ordinary Least Squares (OLS) regression. To address this, a logarithmic transformation was applied to the expenditure variable. The resulting **log-transformed distribution (Figure 2)** is substantially more symmetric, with observations more evenly spread around the center, approximating a normal distribution.

Overall, the logarithmic transformation reduces the influence of extreme values, improves the suitability of the variable for regression analysis, and supports robust statistical inference.

Figure 1: *Histogram of Weekly Household Expenditure (£) showing positive skew and top-coding*

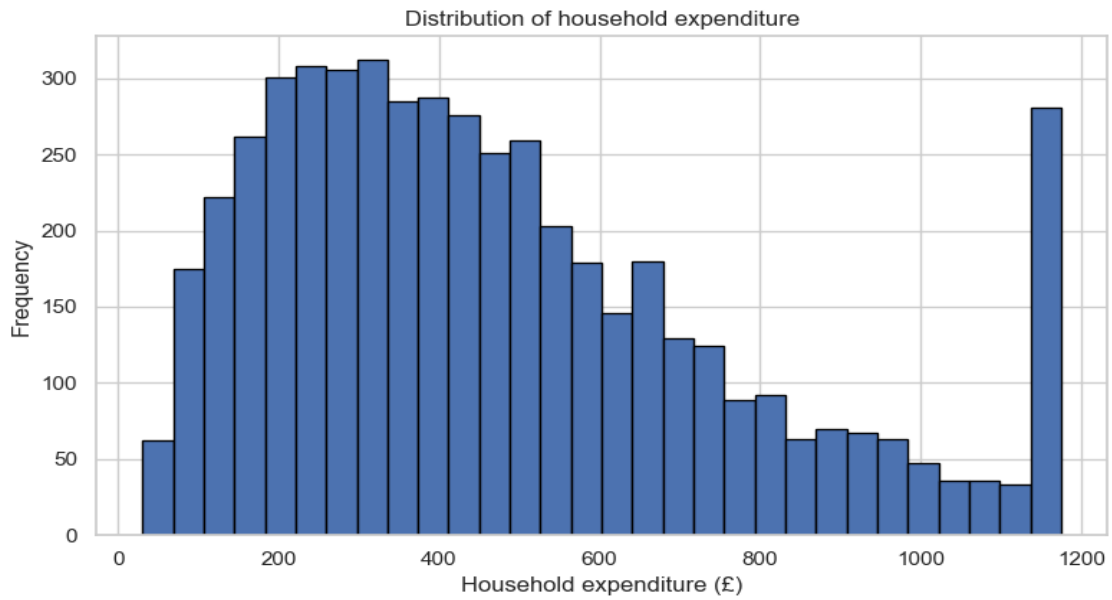
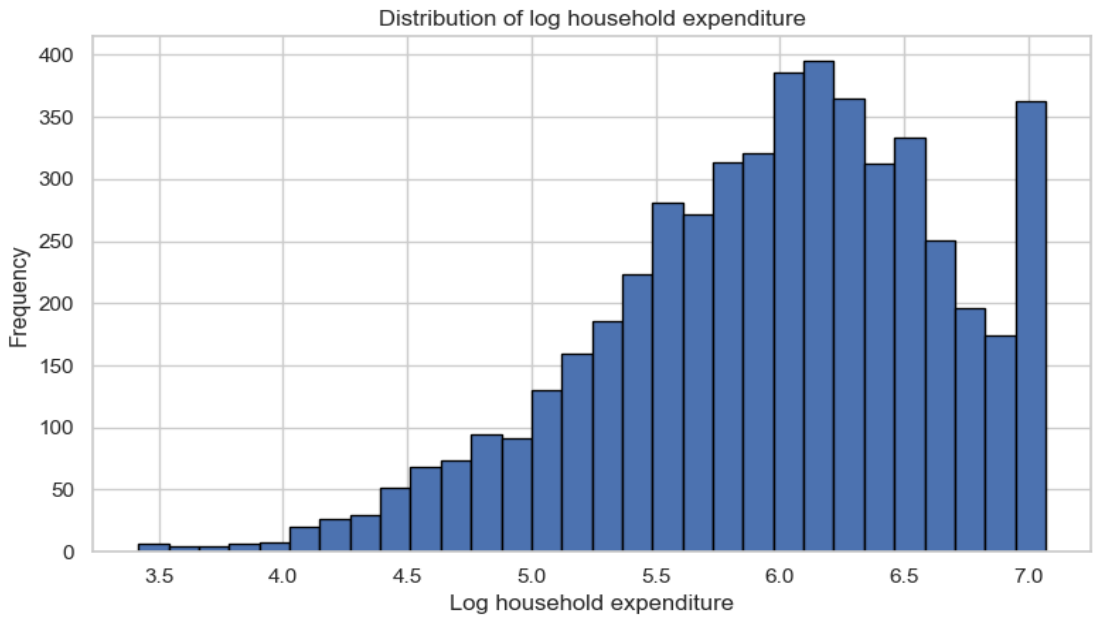


Figure 2: *Histogram of Log-transformed Expenditure showing improved symmetry and normality*



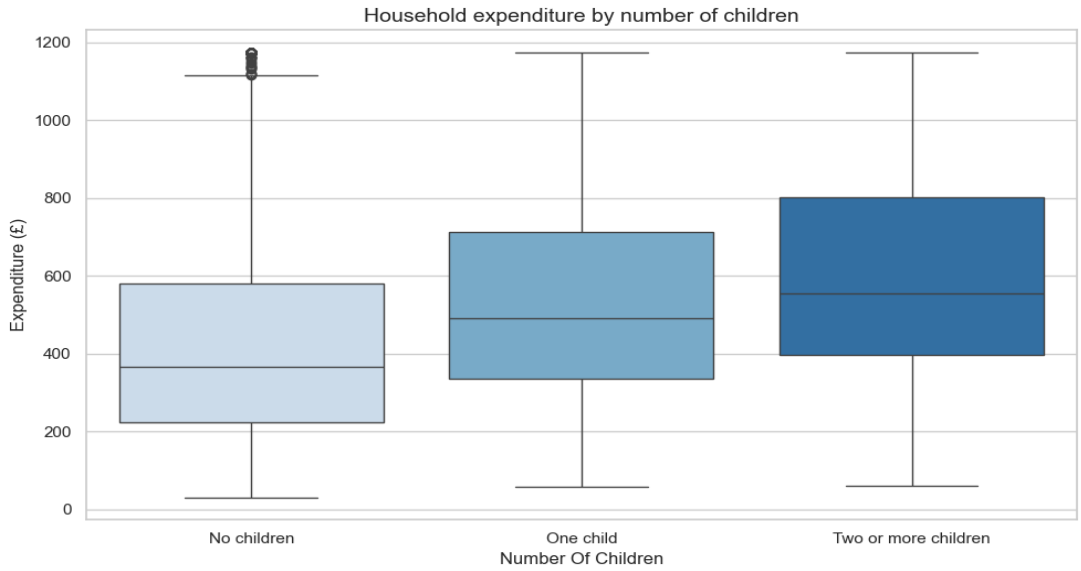
7.2 Expenditure by Tenure type

Figure 3 shows a box plot of **log-transformed household expenditure** by tenure type. The plot reveals distinct differences in expenditure patterns across housing categories. Households in owner-occupied and private rented accommodation tend to have higher expenditures compared to those in public rented housing. The median log expenditure increases progressively from public rented to private rented and is highest for owner-occupied households.

The box plot also highlights variation within each tenure category. Public rented households have a relatively narrow interquartile range, indicating more uniform spending levels, whereas private rented and owner-occupied households show wider distributions, reflecting greater heterogeneity in spending. Numerous lower outliers in the private rented and owner-occupied groups are visible, showing households with unusually low expenditures for their tenure type.

Using the **log-transformed expenditure** reduces the skewness observed in the raw data and provides a clearer comparison across tenure types. These patterns justify the inclusion of tenure type as a categorical predictor in the regression model, as the differences in spending across housing categories are substantial and likely to influence household expenditure.

Figure 3: *Box plot of log-expenditure by tenure type*



7.3 Household Composition: Number of Adults

Figure 4 illustrates the distribution of log-transformed household expenditure by the number of adults in the household.

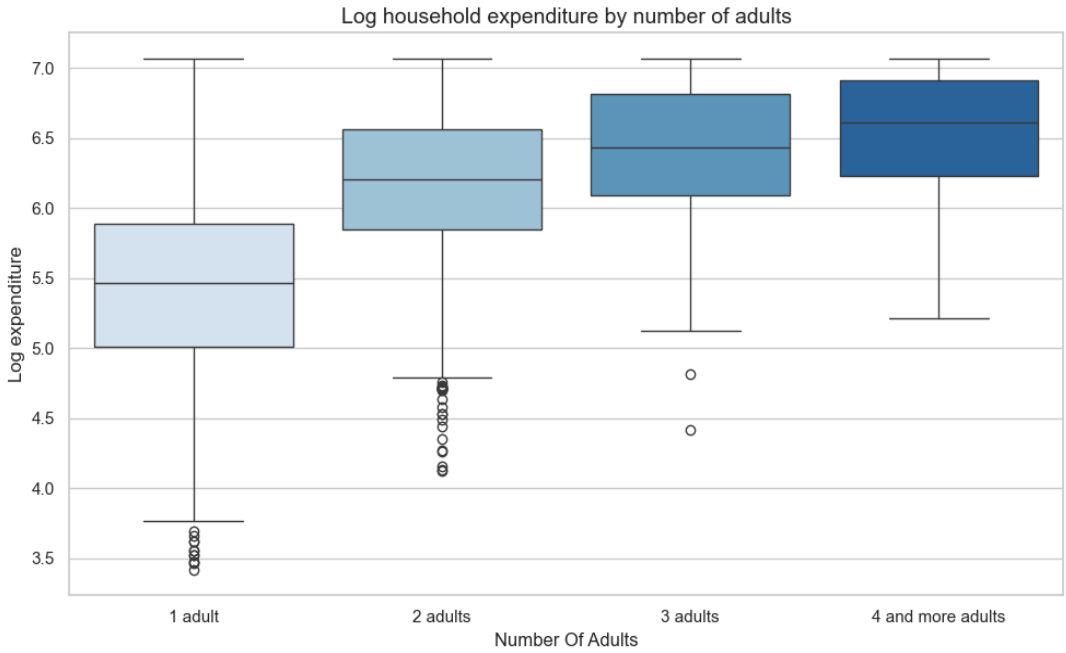
A clear upward trend is visible; as the number of adults increases, both the median log-expenditure and the interquartile range (IQR) shift upward. This confirms a strong positive relationship between household size and total spending.

The most substantial jump in median expenditure occurs between single-adult households and two-adult households. While expenditure continues to rise for three and four-or-more adult households, the gap between the medians appears to narrow slightly, potentially suggesting economies of scale within larger households.

Ceiling Effect (Top-Coding): Notably, the upper whiskers for all categories - especially for households with two or more adults, terminate at approximately the same level (around 7.0 on the log scale). This visually represents the top-coding mentioned in the histogram analysis, where the highest expenditure values are capped at a specific threshold.

Low-Spending Outliers: There is a significant presence of low-spending outliers (indicated by the circles below the whiskers), particularly in one and two-adult households. This highlights the variability in spending habits even within similarly sized households.

Figure 4: *Box plot of log-expenditure by number of adults in the household*



7.4 Household Composition: Number of Children

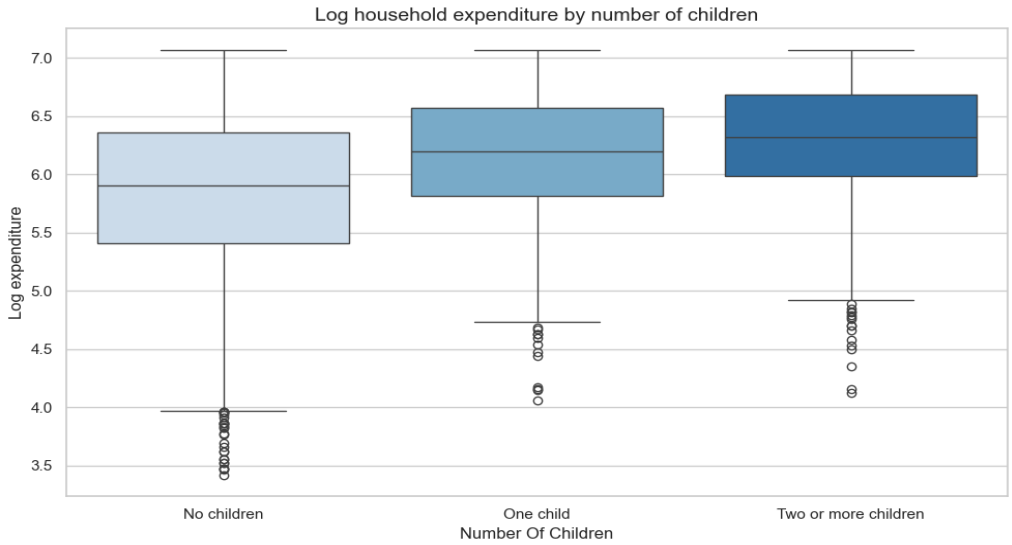
Figure 5 displays the log-transformed household expenditure across three categories of the number of children. There is a discernible increase in median expenditure as children are added to the household. The jump from "No children" to "One child" is more pronounced than the jump from "One child" to "Two or more children," suggesting that while additional children increase total household needs, the marginal cost may decrease slightly for larger families.

Compared to the "Number of Adults" plot, there is a greater degree of overlap between the interquartile ranges (the boxes) of the three categories. This indicates that while children are a statistically significant predictor, they explain less of the total variation in spending than the number of adults does.

A consistent pattern of low-spending outliers is present across all groups. This suggests that even in households with multiple children, some families maintain very low expenditure levels, likely due to differences in disposable income or consumption choices.

As seen in previous visualizations, the maximum log-expenditure values are effectively capped near the 7.0 mark. This further reinforces the observation that high-expenditure households are likely top-coded in the dataset, regardless of family size.

Figure 5: *Box plot of log-expenditure by number of children in the household*



7.5 Occupational Class and Expenditure

In **Figure 6** the box plot displays the distribution of log-transformed household expenditure across five distinct occupational categories. There is a strong, positive correlation between occupational status and household spending, with the highest professional tiers exhibiting significantly higher expenditure levels.

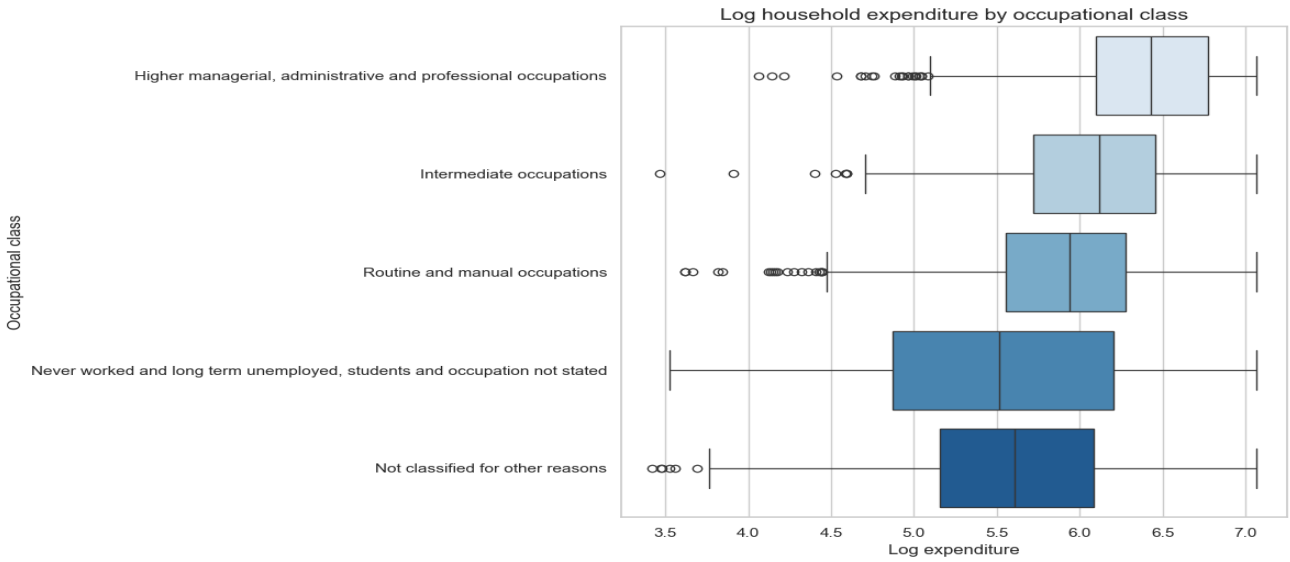
There is a pronounced hierarchical trend in median expenditure. The "Higher managerial, administrative and professional" class shows the highest median log-expenditure (approximately 6.4), followed sequentially by "Intermediate" and "Routine and manual" occupations. This suggests that employment type and, by extension, income level are the primary drivers of household consumption. The classes labeled "Never worked/long term unemployed" and "Not classified" show the lowest medians, reflecting the economic constraints associated with unemployment or non-standard work statuses.

The interquartile ranges (IQRs) vary significantly between classes. The "Never worked and long-term unemployed" category displays a much wider box than the professional classes, indicating a high degree of spending heterogeneity within that group. In contrast, the professional and intermediate classes show tighter clusters around higher spending levels, suggesting more uniform consumption patterns among higher-income households.

The occupational plot shows a dense cluster of low-spending outliers in the top three employment tiers. These outliers represent households in high-status occupations that maintain typically low expenditure levels (log-values between 3.5 and 5.0). The "Never worked" group, while having the lowest median, has fewer individual outliers because its lower "whisker" extends much further down, capturing the extreme low-end of the dataset within the main distribution.

As seen in previous visualizations, all categories reach a ceiling just above the 7.0 mark. Regardless of occupational prestige or income potential, no household exceeds this limit in the plot. This reinforces the hypothesis that the dataset is top-coded, effectively masking the true expenditure levels of the wealthiest households in the higher managerial and professional classes.

Figure 6: *Box plot of log-expenditure by occupational class*



8. STATISTICAL ANALYSIS METHOD

To examine the relationship between household expenditure and the selected explanatory variables, a multiple linear regression model was estimated using Ordinary Least Squares (OLS). This approach is appropriate because the dependent variable (total household expenditure) is continuous, while the predictors are categorical variables describing socio-economic status and household composition.

Multiple regression allows the effect of each explanatory variable on expenditure to be estimated while holding the other variables constant. The model includes occupational class, tenure type, number of adults, and number of children as predictors, so that the association between each factor and household expenditure can be evaluated simultaneously.

All explanatory variables were included in the model as categorical factors. Occupational class and tenure type are naturally categorical variables and were entered using dummy (indicator) variables, allowing each category to be compared with a reference group. The household composition variables (number of adults and number of children) were also treated as categorical factors rather than numeric variables, so that the model could capture possible non-linear differences in expenditure between household types.

The analysis was implemented in Python using the *statsmodels* library. The regression models were estimated using the Ordinary Least Squares (OLS) method, which determines the coefficients by minimizing the sum of squared differences between the observed and predicted values of the dependent variable.

Because household expenditure is typically right-skewed, an additional specification was estimated using the logarithm of expenditure as the dependent variable (log-linear OLS model). The logarithmic transformation reduces the influence of extreme values, improves the distribution of residuals, and allows the coefficients to be interpreted as approximate percentage differences in expenditure. The results from the raw and log-transformed models were compared in order to assess the robustness of the findings and to select the most appropriate specification.

Diagnostic tests and graphical analysis were then performed to evaluate the assumptions of the regression model, including normality of residuals, homoscedasticity, independence of observations, and multicollinearity. The presence of heteroscedasticity was tested using the Breusch–Pagan test. Since the test indicated non-constant variance of the residuals, heteroskedasticity-robust standard errors (HC3) were used. The HC3 estimator is a sandwich covariance estimator that provides valid standard errors, confidence intervals, and p-values even when the assumption of constant variance is violated, ensuring that statistical inference remains reliable.

9. ASSUMPTIONS OF THE ANALYSIS

Multiple linear regression relies on several assumptions that must be considered for the estimated coefficients and statistical inference to be reliable. These assumptions were evaluated using both statistical tests and graphical diagnostic tools.

First, the model assumes a linear relationship between the dependent variable and the explanatory variables. In this study, the predictors were included as categorical factors, allowing the regression model to estimate mean differences in expenditure between groups. In addition, a logarithmic transformation of the expenditure variable was estimated as an alternative specification to improve the functional form and reduce the effect of extreme values.

Second, the observations should be independent. The dataset consists of separate households participating in a national survey, and each household represents a single observation. Therefore, the independence assumption is considered reasonable.

Third, the residuals of the regression model should be approximately normally distributed. Residual normality was evaluated using the Jarque–Bera test and Q–Q plots. The results indicated deviations from normality in the raw expenditure model, reflecting the right-skewed distribution of expenditure. However, the log-transformed model showed an improvement in the distribution of residuals. Given the large sample size (5,144 observations), remaining deviations from normality are unlikely to substantially affect the reliability of the regression estimates.

Fourth, the variance of the residuals should be constant across observations (homoscedasticity). This assumption was tested using the Breusch–Pagan test and examined using residuals-versus-fitted plots. The results indicated the presence of heteroscedasticity, particularly in the raw specification. To address this issue, heteroskedasticity-robust standard errors (HC3) were used to ensure valid statistical inference even when the assumption of constant variance is violated.

Finally, the explanatory variables should not exhibit strong multicollinearity. Multicollinearity was assessed using Variance Inflation Factors (VIF). All VIF values were well below commonly accepted thresholds, indicating that the explanatory variables are not highly correlated and that the regression coefficients can be estimated reliably.

Overall, while some classical assumptions—particularly normality and homoscedasticity—are not perfectly satisfied, the use of a log transformation, a large sample size, and robust standard errors ensures that the regression results remain reliable and suitable for inference.

10. HYPOTHESES

The following hypotheses were tested in the regression analysis to determine whether the explanatory variables are jointly related to household expenditure.

Null hypothesis (H_0):

All slope coefficients in the regression model are equal to zero.

$$H_0: \beta_1 = \beta_2 = \ldots = \beta k = 0$$

This implies that there is no relationship between household expenditure and the explanatory variables (occupational class, tenure type, number of adults, and number of children). In other words, none of the predictors has a statistically significant effect on expenditure.

Alternative hypothesis (H_1):

At least one slope coefficient is different from zero.

H_1: At least one $\beta j \neq 0$

This implies that household expenditure is related to at least one of the explanatory variables included in the model.

The hypotheses were tested using the overall F-test of the regression model, which evaluates whether the explanatory variables jointly improve the prediction of expenditure compared with a model containing only the intercept. In addition, the statistical significance of individual coefficients was assessed using t-tests and the corresponding p-values reported in the regression output.

11. REGRESSION RESULTS

A multiple linear regression model was estimated using Ordinary Least Squares (OLS) to examine the relationship between household expenditure and occupational class, tenure type, number of adults, and number of children. Diagnostic tests indicated the presence of heteroscedasticity; the model was re-estimated using HC3 heteroskedasticity-robust standard errors. Table 3 reports the results for the raw expenditure specification.

The overall regression model is highly statistically significant ($F = 392.2$, $p < 0.001$), indicating that the explanatory variables jointly provide useful information for predicting household expenditure. The model explains approximately 41.2% of the variation in household expenditure ($R^2 = 0.412$, adjusted $R^2 = 0.411$), representing a moderate level of explanatory power for cross-sectional household data. The regression is based on 5,144 observations and includes categorical predictors for occupational class, tenure type, number of adults, and number of children, with reference categories defined for each factor. Robust (HC3) standard errors were used to ensure valid statistical inference in the presence of heteroscedasticity.

The estimated coefficients show clear differences in expenditure across socio-economic groups. Compared with the reference occupational group (higher occupations), households in intermediate occupations spend on average about £128 less per week, while households in

routine and manual occupations spend approximately £195 less. Households classified as never worked, long-term unemployed, students, or not stated also exhibit significantly lower expenditure. All occupational class coefficients are statistically significant at the 1% level ($p < 0.001$), confirming that socio-economic status is strongly associated with household spending.

Housing tenure is also significantly related to expenditure. Relative to households living in public rented accommodation, households in private rented housing spend approximately £104 more per week, while owner-occupied households spend about £133 more. These effects are statistically significant suggesting that housing tenure is an important determinant of household expenditure.

Household composition has the largest effect on predicted spending. Compared with single-adult households, the presence of a second adult increases predicted weekly expenditure by about £194, while households with three adults spend approximately £311 more per week. The presence of children also increases expenditure, although the effect is smaller than that of additional adults. This finding suggests that the number of adults in the household is one of the strongest predictors of total expenditure.

Although the raw expenditure model provides results that are easy to interpret in monetary units (£), diagnostic checks indicated deviations from normality and evidence of heteroscedasticity in the residuals. For this reason, an additional log-linear specification was estimated using the logarithm of expenditure as the dependent variable. The log transformation reduces the influence of extreme values, improves the distribution of residuals, and allows the coefficients to be interpreted approximately as percentage differences in expenditure. The results of the log-linear model are consistent with those of the raw specification, confirming the robustness of the findings. Because the log specification better satisfies the assumptions of the regression model, it was preferred for econometric interpretation.

As a sensitivity check, the analysis was repeated after excluding the 151 households from Northern Ireland, since economic conditions and institutional factors may differ from those in Great Britain. The exclusion did not materially change the estimated coefficients or the overall

conclusions. Therefore, the full sample was retained for the main analysis, and the robustness check is not reported in detail in this document, but the corresponding results are available in the project code.

Overall, the regression results indicate that occupational class, tenure type, number of adults, and number of children are all statistically significant predictors of household expenditure, providing strong evidence against the null hypothesis that these variables are unrelated to spending.

Predictor	Coefficient (£)	Std. Error (HC3)	t-statistic	p-value
Intercept	340.54	10.23	33.28	< 0.001
Occupational Class *(Ref: Higher Managerial)*				
↳ Intermediate	-128.36	10.77	-11.93	< 0.001
↳ Routine and Manual	-194.9	9.14	-21.32	< 0.001
↳ Never worked / Unemployed	-217.52	16.98	-12.81	< 0.001
↳ Not classified	-210.86	9.28	-22.73	< 0.001
Tenure Type *(Ref: Public Rented)*				
↳ Private Rented	103.96	9.57	10.87	< 0.001
↳ Owned	132.51	6.99	18.97	< 0.001
Number of Adults *(Ref: 1 Adult)*				
↳ 2 Adults	193.75	6.6	29.34	< 0.001
↳ 3 Adults	310.75	13.88	22.39	< 0.001
Note: Model F-stat = 392.2; R^2 = 0.412; n = 5,144.				

Table 3: Multiple Regression Results

12. DIAGNOSTIC TESTS AND MODEL VALIDATION

Additional diagnostic tests were conducted to evaluate whether the classical assumptions of the Ordinary Least Squares (OLS) regression model were satisfied. In particular, the normality of the residuals was examined using the Jarque–Bera test, while the presence of heteroscedasticity was assessed using the Breusch–Pagan test. Graphical diagnostics, including Q–Q plots and residuals-versus-fitted plots, were also used to complement the formal statistical tests.

Normality of residuals

The Jarque–Bera test was applied to the residuals of both the raw expenditure model and the log-transformed model. For the raw model, the Jarque–Bera statistic was 821.54 with $p < 0.001$, indicating a strong rejection of the null hypothesis of normality. The residuals were positively skewed and displayed excess kurtosis, suggesting the presence of extreme values.

For the log-transformed model, the Jarque–Bera statistic decreased substantially to 41.93 ($p < 0.001$). Although the null hypothesis of perfect normality is still rejected due to the large sample size, the skewness and kurtosis values are much closer to those expected under a normal distribution. This indicates that the logarithmic transformation improves the distribution of residuals.

The Q–Q plots confirm these results. In the raw model, the residuals deviate from the 45-degree reference line, especially in the upper tail, indicating non-normality. In contrast, the residuals from the log model lie closer to the reference line, showing a better approximation to normality.

Heteroscedasticity

The Breusch–Pagan test was used to evaluate whether the variance of the residuals is constant. For the raw model, the test statistic was highly significant ($p < 0.001$), indicating clear evidence of heteroscedasticity. For the log model, the Breusch–Pagan test also remained significant ($p < 0.001$), although the test statistic was smaller, suggesting that the log transformation reduced but did not completely eliminate heteroscedasticity.

The residuals-versus-fitted plots support these findings. In the raw model, the spread of residuals increases with the fitted values, which is a typical sign of heteroscedasticity. In the log-transformed model, the variance of residuals is more stable, although some heteroscedasticity remains.

Use of robust standard errors

Because heteroscedasticity was detected in both specifications, heteroskedasticity-robust (HC3) standard errors were used for statistical inference. The HC3 covariance estimator provides consistent standard errors even when the assumption of constant variance is violated, ensuring that hypothesis tests and confidence intervals remain valid.

Choice of preferred specifications

The comparison of the diagnostic tests shows that the log-transformed model better satisfies the assumptions of the OLS regression. The transformation reduces skewness, improves the normality of residuals, and leads to more stable variance. For this reason, the log-linear specification is preferred for econometric interpretation, while the raw model is retained for descriptive interpretation in monetary units.

Overall, the diagnostic analysis confirms that the main conclusions of the regression are robust and that the use of the log specification together with HC3 robust standard errors provides reliable statistical inference.

13. INTERPRETATION AND DISCUSSION

The purpose of this study was to investigate whether household expenditure is related to occupational class, tenure type, number of adults, and number of children. The result of the regression analysis indicate a clear and statistically significant relationship between these variables and household spending.

Households with higher occupational classes consistently exhibit higher expenditure, reflecting higher income levels and better economic conditions. Housing tenure is also strongly associated

with expenditure: households in privately rented or owner-occupied homes spend more than those in public rented accommodation, highlighting the role of housing conditions in shaping consumption patterns.

Household composition has the largest impact on expenditure. Expenditure increases substantially with the number of adults in the household, which is expected as more adults lead to higher consumption and living costs. Similarly, households with children spend more, with the effect growing as the number of children increases, although the marginal impact of children is smaller than that of additional adults.

These findings are consistent with economic theory, which predicts that both socio-economic status and household size influence spending behavior. These findings provide useful insights into differences in living standards between households and have implications for policymakers concerned with the cost of living, inequality, and social welfare. The log-linear specification further confirms these relationships, offering a more reliable framework for statistical inference due to improved residual properties and stability.

14. CONCLUSION

This case study analyzed the determinants of household expenditure using data from the 2013 Living Costs and Food Survey teaching dataset. Multiple linear regression models were estimated to examine the effects of occupational class, tenure type, number of adults, and number of children on total household expenditure.

The analysis revealed that all included variables are statistically significant predictors of expenditure. Households with higher occupational status, owned or privately rented housing, more adults, and more children tend to have higher spending levels. The preferred log-linear model explained approximately 41% of the variation in expenditure, indicating a moderate level of explanatory power for cross-sectional household data.

Overall, the findings highlight the combined importance of socio-economic status and household composition in shaping household expenditure. This study demonstrates how survey data can be used to identify patterns in household consumption and provides a basis for informed economic policy, targeted social programs, and further research into household living standards.

15. REFERENCES

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